

CHERDYN, Russian Federation

WMO: 239140

Lat: 60.400N

Lon: 56.517E

Elev: 207

StdP: 98.87

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.3	-28.9	-34.8	0.1	-32.2	-31.4	0.2	-28.8	9.3	-9.5	8.3	-8.9	1.6	320	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	28.3	19.9	26.3	18.9	24.4	17.5	20.7	26.8	19.6	24.9	18.6	23.2	2.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.7	23.7	17.5	12.9	22.8	16.6	12.1	21.6	60.7	26.9	57.0	24.9	53.4	23.3	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.3	6.4	DB	-34.9	30.5	4.2	1.5	-37.9	31.6	-40.4	32.5	-42.8	33.4	-45.8	34.5	(4)
(5)			WB	-34.9	22.3	4.2	1.5	-38.0	23.4	-40.4	24.3	-42.8	25.2	-45.8	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-14.7	-13.2	-5.2	2.1	9.3	15.1	17.9	14.3	8.7	1.5	-7.1	-12.5	(6)	
	DBStd	12.65	7.63	7.27	4.88	4.90	5.57	4.65	4.55	4.33	4.24	4.58	6.66	8.18	(7)	
	HDD10.0	3783	766	649	472	241	84	10	2	8	73	266	513	698	(8)	
	HDD18.3	6263	1024	883	730	488	284	118	64	141	289	521	763	956	(9)	
	CDD10.0	650	0	0	0	3	62	162	247	141	34	2	0	0	(10)	
	CDD18.3	89	0	0	0	0	3	21	51	14	1	0	0	0	(11)	
	CDH23.3	626	0	0	0	0	26	143	358	96	3	0	0	0	(12)	
	CDH26.7	132	0	0	0	0	2	27	79	23	0	0	0	0	(13)	
	Wind	WSAvg	2.9	2.9	3.1	3.4	3.1	2.9	2.7	2.1	2.4	2.8	3.2	3.0	2.8	(14)
(15) Precipitation	PrecAvg	712	51	35	35	46	53	67	84	74	66	72	68	60	(15)	
	PrecMax	964	94	69	105	102	125	170	189	163	123	147	157	113	(16)	
	PrecMin	480	8	1	2	15	11	23	15	22	17	14	3	6	(17)	
	PrecStd	112	19	15	21	23	26	38	39	33	28	28	29	23	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	0.6	7.1	19.0	26.3	29.5	30.4	29.7	22.6	15.1	7.0	1.7	(19)	
		MCWB	-0.4	-0.1	2.4	10.2	15.6	20.4	20.7	20.2	15.8	9.9	6.5	1.2	(20)	
	2%	DB	-1.2	-0.2	4.0	14.3	23.5	27.0	28.9	25.8	18.9	12.0	3.7	0.4	(21)	
		MCWB	-1.6	-0.7	0.8	7.9	14.1	18.8	20.3	19.1	14.0	9.1	3.1	0.1	(22)	
	5%	DB	-2.9	-1.5	2.2	11.2	21.0	24.8	27.2	23.2	16.7	9.7	1.6	-0.6	(23)	
		MCWB	-3.2	-1.9	0.0	5.8	12.7	17.2	19.6	17.5	12.9	7.8	1.2	-0.9	(24)	
	10%	DB	-4.7	-3.2	0.9	9.0	18.4	22.6	25.3	20.9	14.7	7.8	0.3	-2.4	(25)	
		MCWB	-5.0	-3.7	-0.6	4.7	11.6	16.2	18.8	16.4	11.9	6.6	0.0	-2.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	0.2	3.2	11.0	17.1	21.3	22.3	21.1	16.9	11.4	6.2	1.4	(27)	
		MCDB	-0.1	0.4	5.8	16.2	23.2	28.0	28.6	27.6	21.2	13.2	6.8	1.5	(28)	
	2%	WB	-1.5	-0.6	1.6	8.4	15.3	19.6	21.2	19.6	14.8	9.5	3.1	0.1	(29)	
		MCDB	-1.2	-0.3	3.2	13.2	21.1	25.1	27.3	25.1	17.9	11.5	3.5	0.3	(30)	
	5%	WB	-3.2	-1.9	0.5	6.6	13.8	18.2	20.3	18.1	13.4	8.0	1.2	-1.0	(31)	
		MCDB	-2.9	-1.5	1.7	10.2	19.2	23.2	25.8	22.1	15.8	9.5	1.5	-0.7	(32)	
	10%	WB	-5.0	-3.6	-0.4	5.1	12.3	17.0	19.3	16.9	12.3	6.6	0.1	-2.8	(33)	
		MCDB	-4.7	-3.2	0.8	8.4	17.7	21.5	24.2	20.1	14.3	7.9	0.3	-2.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	5.9	6.7	7.5	9.0	8.6	8.6	7.3	6.2	4.1	4.1	5.1	(35)	
		MCDBR	5.4	4.9	7.2	10.7	12.0	11.0	10.7	10.1	9.0	6.8	3.8	4.2	(36)	
	5% WB	MCWBR	5.5	4.9	5.1	6.4	5.9	5.3	4.8	4.6	4.9	4.5	3.8	4.4	(37)	
		MCWBR	5.6	5.0	6.0	9.7	10.8	9.9	9.3	7.9	6.2	3.7	3.7	4.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.272	0.305	0.349	0.375	0.374	0.372	0.401	0.396	0.357	0.326	0.280	0.257	(40)	
		taud	1.969	2.012	1.982	2.074	2.353	2.415	2.332	2.327	2.396	2.301	2.057	1.921	(41)	
	Edn at Noon	Edn at Noon	434	649	744	809	851	859	818	790	759	646	428	287	(42)	
		Edh at Noon	48	84	121	133	109	105	111	104	82	64	44	33	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.35	1.11	2.27	3.62	4.77	5.08	5.25	3.67	2.07	0.83	0.32	0.18	(44)	
	RadStd	RadStd	0.05	0.24	0.32	0.54	0.52	0.47	0.52	0.45	0.23	0.11	0.08	0.03	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page